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Original Article

Comparison of experimental, analytical and simulation results for hot rolling of S275JR quality steel

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ABSTRACT

Hot rolling allows the material to be shaped in successive steps in different steps since the material is at a temperature above 0.5 Tm which is the recrystallization temperature and gives the possibility of deformation with large deformation rates. In the industry, direct applications show negative effects on time and cost. Practical processes by experimenting can be completed in short periods in some symmetrical profile section products and long periods in some asymmetric profile section products. Finite element method (FEM) can be applied in order to predict the results obtained by simulating the designed calibrations in the computer environment before testing in the rolling mill. In this study, the production of HEA 240 profile in S275JR quality is experimentally made in 21 passes and the geometric dimensions of the first three passes of this structure was compared with 3D FEM analysis results. The comparison results are compatible and high similarity ratio as 95–99.1% was found. Following to this, the analytically calculated dimensions of calibrated IPE 140 profile in S275JR quality was compared by the verified 3D FEM analysis. Considering the first 5 passes of 15 passes for saving time, the simulation and analytical results are in close relationship less than 5% deviation ratio. It is possible to say that this simulation, which has small acceptable differences, has a significant effect on minimizing losses in production processes and achieving production targets.

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1. Introduction

Rolling is a plastic forming method in which the desired shape is obtained by pulling the metal material to be rolled with the effect of friction force between two rollers rotating in opposite direction to each other and obtaining the desired shape [1,2].

The rolling method is divided into two groups, cold rolling and hot rolling, depending on the forming process temperature. Hot rolling is one of the most important industrial processes for the finished or semi-finished bulk production [3–6].

Today, global competition forces companies to find out how they can produce their products with higher quality and lower costs. Simulation is defined as simulating real-world processes and can be used to illustrate the real effects of alternative conditions and behavior patterns. Simulations can also be performed on systems that are inaccessible, dangerous,

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impossible to operate, or are currently under construction or not yet built [7–9]. Generally, the main reason for simulation is low cost. Thanks to the simulation, it is possible to quickly and inexpensively determine whether a job is correct or not. In this way, it will allow conducting experiments without wasting high costs and time. Furthermore, it will contribute to increasing the speed of existing production processes and product quality.

LI Chang-Sheng et al. [10] studied the three-dimensional (3D) temperature area of the roll, which worked during the hot rolling of the strip profile using FEM, based on the heat conduction equation and analyzed the area of temperature change and thermal stress occurring on the surface of the roll. They stated that the highest roller surface temperature was 593 °C and the difference between the maximum-minimum thermal stresses of the roller surface was 145.7 MPa. A. R. Shahani et al. [11] focused on the thermomechanical analysis of AA5083 aluminum alloy using FEM in the hot rolling process. In the analysis of the process, the main variables rolling process characterization were formed by a mathematical and geometric model by subtracting the areas of temperature distribution, tension, elongation and elongation ratio in the roll and billet in a parametric form. Iankov simulated the wire rolling process in two and three dimensions. The simulation results were compared with the experimental data [12]. Xiong Shangwu et al. [13] performed thermal coupling analysis with the three-dimensional rigid-plastic Finite Element Method (FEM) to simulate the vertical-horizontal rolling process in hot strip profile rolling. They found that there were small acceptable differences between the results of the theoretical model they created and the experimental data. Rout et al. [14] analyzed the material flow, spread and elongation velocities on four side surfaces using finite element modeling (FEM) for hot cross-rolling of 304 stainless steel plates. They stated that the edge profiles prescribed by FEM are in agreement with the measured profiles but that the corner profiles deviate slightly from the experimentally obtained profiles. Milen et al. [15] conducted two-pass oval round rolling and 6-pass equal edge gusset rolling analysis. In the study based on dislocation theory, they used FE-based SortRoll computer software and analyzed the material flows in each pass and explained that they were in harmony with the actual values. Using FEM-based FORGE software, Nalawade et al. [16] simulated thermo-mechanical extrusion of a micro-alloyed 38MnVS6 grade steel bloom. They examined the effects of rolling parameters on deformation behavior and experimentally confirmed the estimates of rolling load, torque, temperature distribution, stress distribution, grain size and structure phase in the rolling phase. Hanoğlu et al. [17] studied the analysis of rolling of an asymmetric profile using FEM. They found that the results obtained by using a certain number of slicing models perpendicular to the rolling direction were in good agreement with the actual production results of the asymmetric section profile they analyzed. Wang et al. developed a kinetically static softening model for inter-pass softening based on the results of hot rolling experiments. They applied an explicit static softening model in the hot rolling of a round bar with a three-dimensional finite element model and simulated static softening. They emphasized that the static softening took place in the first 2 s at the beginning and that the temperature was more effective than the rolling speed [18]. Braga et al. examined the recrystallization formation depending on the temperature in hot rolling and examined this formation with optical microscopy and electron backscatter diffraction (EBSD). They suggested that the first passes at high temperatures for required recrystallization, lower temperatures should be applied at the last passes for required recrystallization and work hardening [19]

Factors such as customer demand, developing technology and global competition force companies to find out how they can produce the product they produce faster, with higher quality and a lower cost. In systems that are physically large, technologically complex, with a wide range of product groups and a large number of parameters and operations, the new product deployment process is very costly and time-consuming. Today, simulation programs are an important alternative to see the real effects of alternative conditions and behavior patterns. It can be said that simulation has important contributions considering production costs and times. Besides, in addition to expensive experimental procedures, the production of systems that are not in production or that are not available can be predicted in this way. This study aims to contribute to the understanding of the roller caliber design, which has an important place in the widely used rolling process, to achieve the real results with the simulation method, and to prevent the loss of time and cost spent on trial production in new product designs. The first step of this study is to compare the sample production results of the HEA 240 structural profile to be formed by coarse rolling (3 passes) of the beam blank with 280 × 360 mm dimension and S 275 JR quality and the results obtained by using 3D FE analysis with the same parameter comprising. Secondly, to minimize the time and cost inconveniences in production processes, 5-pass rough rolling of IPE 140 structure profile with different calibrated design and cross-sectional dimensions was performed by using the proven 3D FE analysis and the production was not performed but analytical calculated results were compared with FE results.

2. Experimental, simulational and analytical method

The study includes a 3D Finite Element (FE) analysis for two different products. Firstly, FE analysis of the first 3 passes of HEA 240 profile obtained in 21 passes with actual production was performed and the accuracy of the simulation was proved by the results of the sample obtained from the actual production. Secondly, FE analysis of the first 5 passes of the IPE 140 profile, which was designed by analytical calculations and to be obtained at 15 passes, as compared with the calculated design results and the similarity rates and compatibility were shown. Fig. 1 shows the experimental and simulation method schematically.

2.1. *Material and equipment*

In the production of the HEA 240 profile, S275JR quality profile draft (Beam Blank) is used. 1200 °C annealed profile draft, alloy steel rollers are used in the reversible rolling table after the

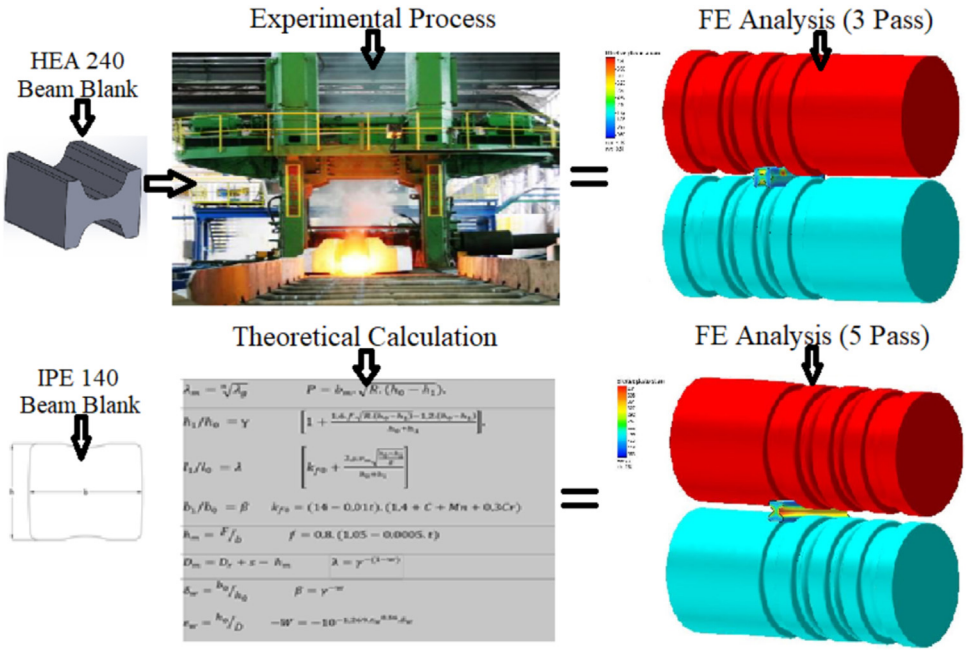


Fig. 1 – Schematic diagram.

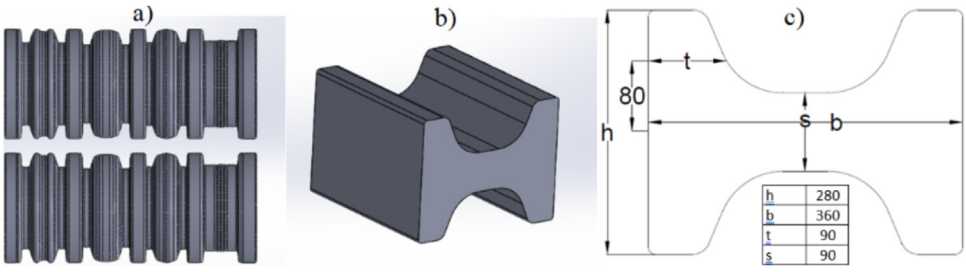


Fig. 2 – 3-pass rough rolling process, a) rolls, b) beam blank, c) beam blank dimensions.

3-pass rough rolling process was sampled. The roller set and semi-finished products used in production are shown in Fig. 2.

The calibrated design of the IPE 140 profile is made of S275JR billet (150 × 150) as a semi-finished product. The caliber design was made from a billet annealed to 1200 °C and made according to a reversible rolling stand using alloy steel rollers. The calibrated roller set and the semi-finished product is shown in Fig. 3. The material properties of the semifinished product are given in Table 1–3.

2.2. *3D finite element (FE) analysis*

In this study, two different products have been analyzed and Simufact Forming, a commercial finite element software, has been used for simulations. In 3D FE analysis, rough rolling simulation of the first 3 passes of HEA 240 product (Fig. 4) produced from profile draft (Beam Blank) and rough rolling simulation of the first 5 passes of IPE 140 product (Fig. 5) calibrated design starting from 150 × 150 billet were performed. Fig. 4 and 5 show the sequence of the simulations for HEA 240 and IPE 140 profiles.

The apparatus used in the analysis consists an of upper and lower roll, beam blank and billet, pusher and gripper geometries. Roller, beam blank and billet geometries were created as a solid model in the CAD program and then imported for analysis. The rolls used in the analyzes were defined as non-conductive and rigid molds. In the analysis, the material quality of the beam blank and the log were chosen as St44-3_h (S275J0) quality in the program library due to the absence of the S275JR quality steel. St44-3 (S275J0) material quality has the same characteristics as S275JR. St44-3 (S275J0) material properties are given in Table 3–5. The heating process was applied at 1200 °C and the geometries obtained from this process were used for the first pass. In the other passes, the geometries formed by the previous pass are used for each.

Hexahedral was selected as the mesh type in the analysis and the element length used in each pass is given in Table 6. The analysis was carried out on a personal computer. Therefore, the determination of element size was influenced by the analysis process and the surface quality obtained from the analysis. The lengthening of the analysis process is very small element sizes has been observed in several trials where surface sensitivity decreases in very large element sizes. In Fig. 6,

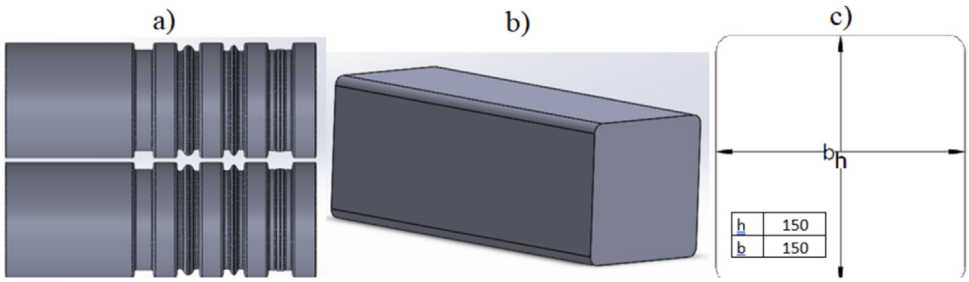


Fig. 3 – 5-pass rough rolling process, a) rolls, b) billet, c) billet dimensions.

Table 1 – Chemical composition for grade S275JR Beam Blank and Billet [20].

Symbol		Nominal thickness of product (mm) for %C (max)			% Mn (max)	% P (max)	% S (max)	% N (max)
TS EN 10025-2	DIN 17100	≤ 16	> 16, 40 ≤	> 40				
S275JR	St44-2	0,21	0,21	0,22	1,5	0,035	0,035	0,012

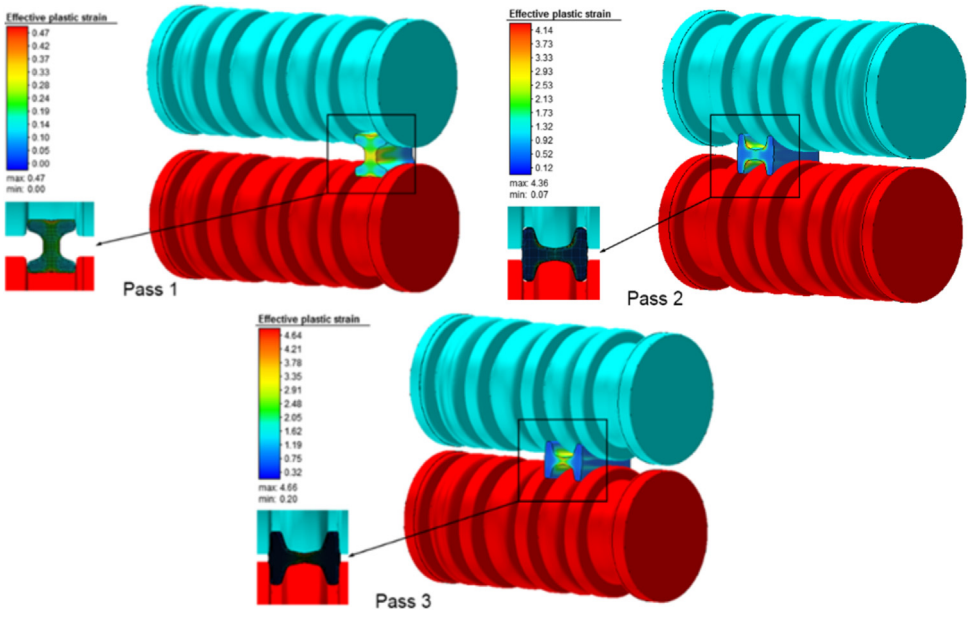


Fig. 4 – Schematic representation of the FE simulation setup for HEA 240.

the mesh structure applied in the analysis of both products is given as an example.

In the analysis, Tabular Motion was selected as the press type. In addition, the rotation of the roller (rpm) is defined depending on the time. Some parameters used in the simulation of the HEA 240 and IPE 140 profiles are given in Table 7. In addition, the number of steps is determined in the analysis depending on the mesh element size and step numbers for each pass are given in Table 8.

2.3. *Analytical analysis of roll pass design*

The caliber design of the IPE 140 profile, which can be produced from 150 × 150 semi-finished products, has been analytically calculated. In hot rolling, mathematical formulas are used to calculate parameters such as elongation, expansion, crushing, the strength of the extruded material at rolling temperature, coefficient of friction and rolling force (Table 9).

In the whole system, an average degree of elongation λm is accepted, provided that it applies to all deformations, or if this degree of elongation is calculated, n is calculated using Eqn 1. λg gives the total degree of elongation. γ, λ and β are the degrees of crushing, elongation and expansion respectively. The roll contact surfaces of the rolled materials may not always have a smooth surface. That is, there may be diagonal, oval, round or asymmetric profile section surfaces. In this case, the cross-sectional height is indicated by the mean height and expressed in hm [Eq 5]. b is the largest width of the cross-sectional area. If the diameter values at the points where the roller contacts the material are different across the material cross-section width, the roller diameter is indicated by the average diameter (Dm) [Eq 6]. Volume stability is

Table 2 – Mechanical properties of grade S275JR steel at ambient temperature [20].

		Yield strength (min), MPa									Tensile strength, MPa				
		Nominal thickness, mm									Nominal thickness, mm				
TS EN 10025-2	DIN 17100	≤16	>16 ≤ 40	>40 ≤ 63	>63 ≤ 80	>80 ≤ 100	>100 ≤ 150	>150 ≤ 200	>200 ≤ 250	>250 ≤ 400	<3	≥3 ≤ 100	>100 ≤ 150	>150 ≤ 250	>250 ≤ 400
S275JR	St44-2	275	265	255	255	235	225	215	205	–	430-580	410-560	400-540	380-540	–

Table 3 – Material parameters for Beam Blank and Billet [21,22].

Material grade	Density (kg/dm^3)	Poisson's ratio	Elastic Modulus (MPa)	Thermal conductivity (W/m.K)	Specific heat (J/kg°K)	Thermal expanion (10^{-6} °K^{-1})
Unalloyed structural steels	7.85	0.3	210	48–58	490	12

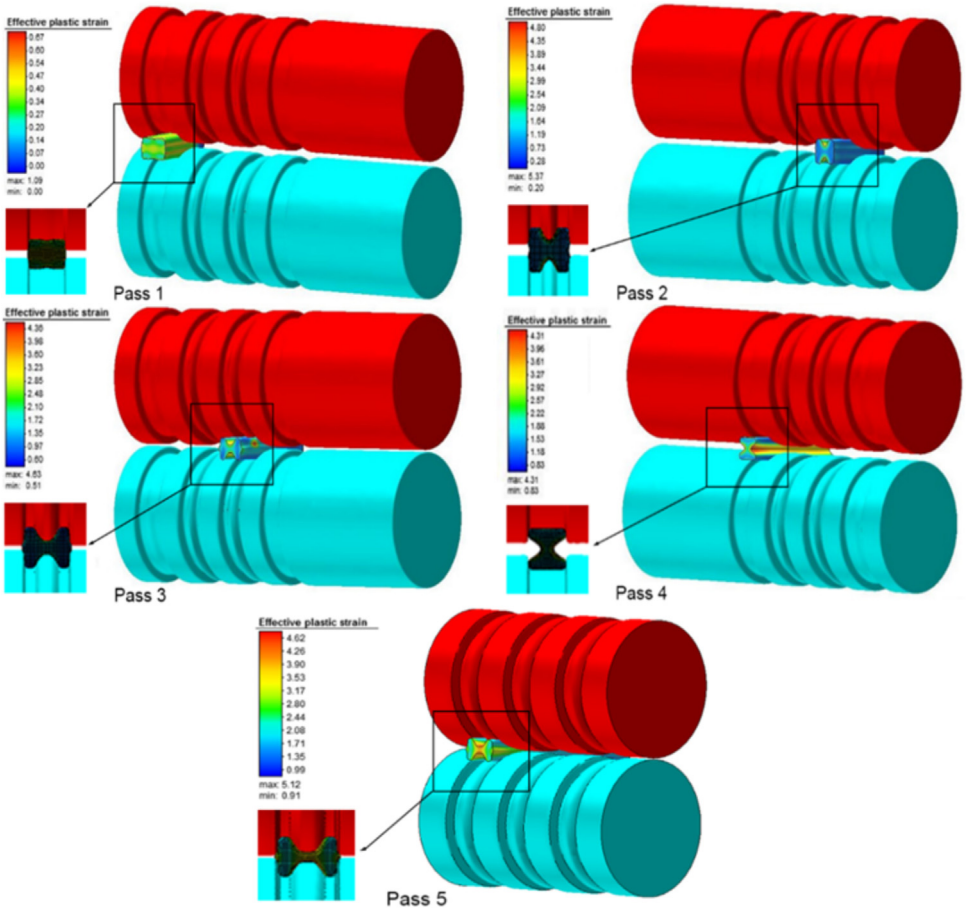


Fig. 5 – Schematic representation of the FE simulation setup for IPE 140.

Table 4 – Chemical composition for grade St44-3(S275J0) [20].

Symbol		Nominal thickness of product (mm) for %C (max)			% Mn (max)	% P (max)	% S (max)	% N (max)
TS EN 10025-2	DIN 17100	≤ 16	> 16, 40 ≤	> 40				
S275J0	St44-3	0,18	0,18	0,18	1,5	0,03	0,03	0,012

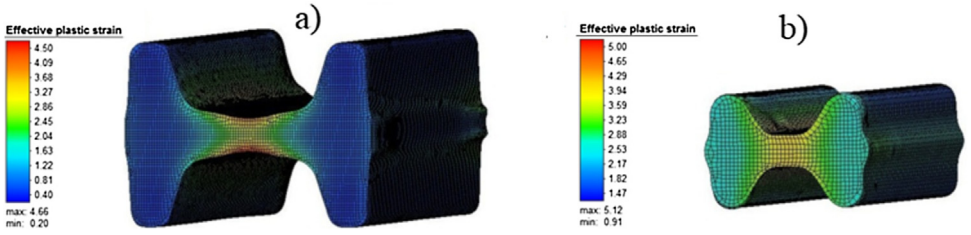


Fig. 6 – The mesh structure of the workpiece a) HEA 240 3. Pass b) IPE 140 5. Pass.

Table 5 – Mechanical properties of grade S275JR steel at ambient temperature [20].

		Yield strength (min), MPa									Tensile strength, MPa				
		Nominal thickness, mm									Nominal thickness, mm				
TS EN 10025-2	DIN 17100	≤16	>16 ≤ 40	>40 ≤ 63	>63 ≤ 80	>80 ≤ 100	>100 ≤ 150	>150 ≤ 200	>200 ≤ 250	>250 ≤ 400	<3	≥3 ≤ 100	>100 ≤ 150	>150 ≤ 250	>250 ≤ 400
S275J0	St44-3	275	265	255	255	235	225	215	205	–	430-580	410-560	400-540	380-540	–

Table 6 – Elements sizes used in analysis.

	Pass 1		Pass 2		Pass 3		Pass 4		Pass 5	
	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140
Mesh (mm)	8	4	5	4	4	3,5	–	3,5	–	3,5

Table 7 – Parameters for rotation of roll.

	Pass 1		Pass 2		Pass 3		Pass 4		Pass 5	
	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140
Rotation (rpm)	66	74	47	73	70	73	–	73	–	55
Time (sec)	0,35	0,35	0,5	0,7	0,4	0,7	–	0,7	–	1,0

Table 8 – Steps value.

	Pass 1		Pass 2		Pass 3		Pass 4		Pass 5	
	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140	HEA 240	IPE 140
Step	330	670	670	700	1000	700	–	750	–	800

Table 9 – Analytical Formulas [1].

Steps	Definitions		Formulas	
1	Number of pass		$\lambda_m = \sqrt[n]{\lambda_g}$	(1)
2	Ratio	Crushing	$h_1/h_0 = \gamma$	(2) (3) (4)
		Elongation	$l_1/l_0 = \lambda$	
		Expansion	$b_1/b_0 = \beta$	
3	Average material height		$h_m = F/b$	(5)
4	Average roll diameter		$D_m = D_r + s - h_m$	(6)
5	Wusatowski formula	Edge ratio	$\delta_w = b_0/h_0$	(7)(8)(9)(10)(11)
		Diameter ratio	$\varepsilon_w = h_0/D$	
		Wusatowski coefficient	$-W = -10^{-1,269.\varepsilon_w{}^{0,56}.\delta_w}$	
		Elongation ratio	$\beta = \gamma^{-w}$	
		Expansion ratio	$\lambda = \gamma^{-(1-w)}$	
6	Ekelund rolling force formula	Rolling force	$P = b_m.\sqrt{R.(h_0 - h_1)}.$	(12) (13) (14)
		Friction coefficient	$\left[1 + \frac{1,6.f.\sqrt{R.(h_0-h_1)}-1,2.(h_0-h_1)}{h_0+h_1}\right].$	
		Material forming strength at forming temperature	$\left[k_{f0} + \frac{2.\mu.v_m\sqrt{\frac{h_0-h_1}{R}}}{h_0+h_1}\right]$	
			$f = 0,8.(1,05 - 0,0005.t)$	
			$k_{f0} =$	
			$(14 - 0,01t).(1,4 + C + Mn + 0,3Cr)$	

Table 10 – Numerical parameters calculated for all passes.

	1. Pass	2. Pass	3. Pass	4. Pass	5. Pass
Roll Gap (mm)	30	31	11	36	6
Dort (mm)	647	655	662	647	542
Speed (m/sn)	2,5	2,5	2,5	2,5	1,4
Revoluation (dev/dak)	74	73	72	74	50
γ	0,800	0,706	0,766	0,864	0,654
β	1,0634	1,1373	1,0748	1,0532	1,1050
λ	1,175	1,245	1,215	1,098	1,383
δ	1,00	0,75	1,21	0,82	1,29
ε	0,232	0,243	0,170	0,226	0,185
W	0,2756	0,3693	0,2701	0,3553	0,2357
f	0,36	0,70	0,72	0,40	0,41
kf0 (kg/mm$\hat{2}$)	5,6	6,4	7,3	8,1	9,0
Force (kN)	1020	1138	1124	320	1109

Table 11 – Results of experimental and simulative for HEA 240.

Dimension (mm)	Experiment	Simulative	% Deviation
h	256	260	1,6
b	352	355	0,9
s	40	42	5,0
t	73	71	2,7
Area (mm^2)	49485	50033	1,1

considered to be the basis in the rolling process. The height and width before rolling will change after the rolling so that the final volume remains constant. In other words, height will decrease as a result of rolling and width and length will increase. Wusatowski [1] has developed the expansion formula [Eq 7–11] which provides volume stability, taking into account the rolled cross-section and roll diameter. During the rolling process, a force must be applied to overcome the yield strength of the material to form the material in plastic. It is also known that frictions occur between the material and the roller contact surfaces and that additional force is required to overcome these frictions. As a result, the force to be applied for the extrusion of the materials must correspond to the total deformation strength, i.e., the sum of the yield strength at the rolling temperature of the material and the additional strength value caused by friction [Eq 12]. The coefficient of friction (f) is calculated by Eq 12. With the help of Eq 14, the deformation resistance of S275 JR structural steel at rolling temperature is obtained. The calculation results for the first 5 passes in the caliber design stage are given in Table 10.

Table 12 – HEA 240 rolling data obtained from experimental data and simulation.

	Pass 1		Pass 2		Pass 3	
	Exp	Sim	Exp	Sim	Exp	Sim
Force (kN)	1450	1300	3000	3500	3500	4100
% Deviation	10,3		16,6		17,1	

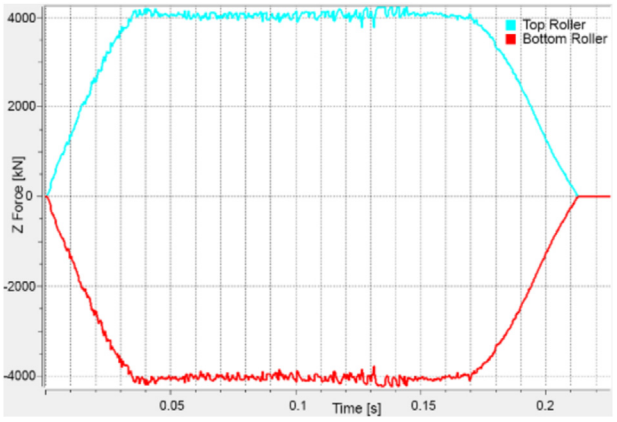


Fig. 8 – Simulational force result for HEA 240 in 3rd pass.

3. Results and discussions

3.1. *Comparison of experimental and FE analysis results (HEA 240)*

The cross-sectional geometry of the HEA 240 profile according to the analysis results of the 3rd pass and the cross-sectional

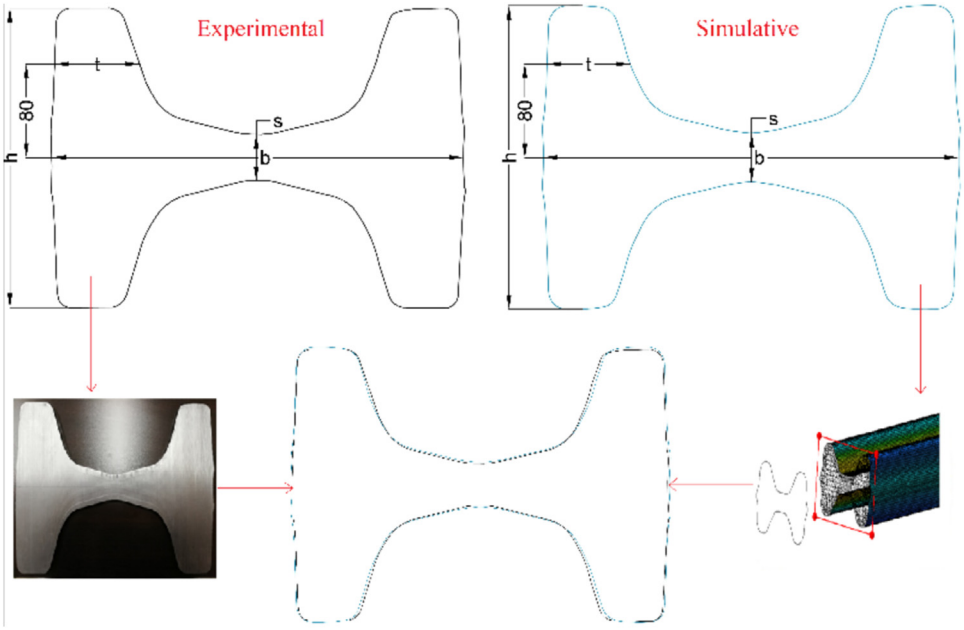


Fig. 7 – Comparasion between experimental and simulative for HEA 240.

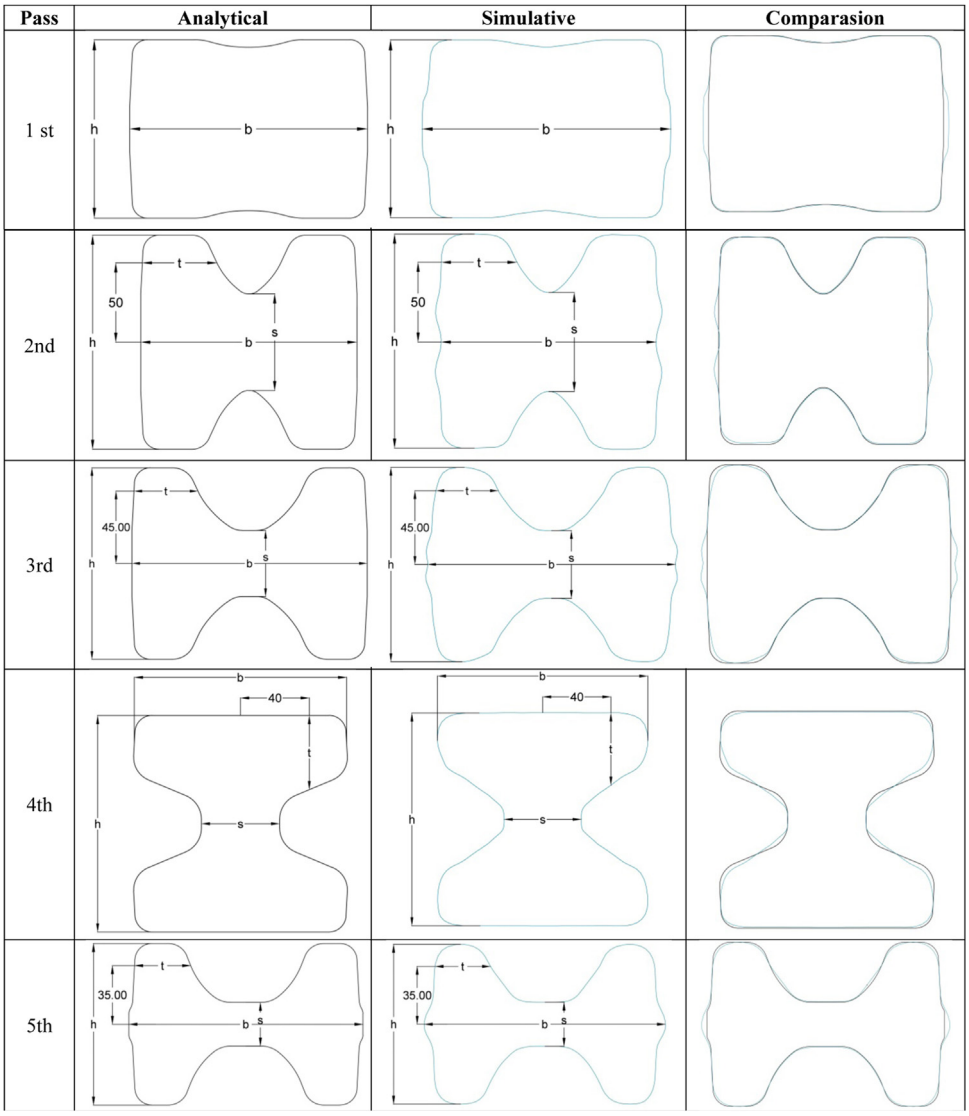


Fig. 9 – Comparasion between calculated and simulative for IPE 140.

geometry of the sample is given in Fig. 7. Measurements are the basic measurements of cross-sectional geometry and the values are given in Table 11.

The geometry and measurements obtained from the simulation show 95% to 99.1% similarity to the geometry and measurements in the sample taken from the actual production. This similarity shows that it is possible to estimate the actual production data with simulation results [16,23]. The closest relationship was obtained with a size of 0.9%. Furthermore, it may be considered that in the more accurate analysis by reducing the mesh (dimension of the element) [24], the differences in dimensions between the two sections will be further reduced, for example by the dimensions indicated by s and t.

In Table 12, the rolling forces obtained as rolling data during production and the rolling forces obtained as a result of the analysis (Fig. 8) were compared. The difference between the rolling forces ranges from 10% to 17%. It should be considered that the difference between the actual production conditions and the determination of the actual production conditions (friction coefficient, the rolling temperature of the material, etc.) will decrease. Nevertheless, the difference between 10% and 17% is acceptable to a certain extent in hot rolling processes. Because the sample moves forward with the rotation of the upper and lower rollers in the experimental production, while it moves with the thrust force given in a certain direction in the simulations. This situation is thought to be effective in the force deviations. Also, it can be said that the manufacturing method may affect this difference in the case of rough rolling. For this reason, the final product is focused on comparing the experimental simulation results.

3.2. *Comparison of experimental and FE analysis results (IPE 140)*

The calibrated cross-sectional geometries designed in Fig. 9 are compared with the cross-sectional geometries obtained by simulation of each sheet. The dimensions shown in section

Table 13 – IPE 140 Cross section measurements calculated at 5 passes and as a result of simulation.

Dimension (mm)	Pass 1		Pass 2		Pass 3		Pass 4		Pass 5	
	Anal	Sim	Anal	Sim	Anal	Sim	Anal	Sim	Anal	Sim
h	120	119,8	135	134,2	120	120	126	125	96,5	96,7
% Deviation	0,2		0,6		0,0		0,8		0,2	
b	160	167,2	136	134,5	145	150	123,5	123,5	139,5	145,5
% Deviation	4,5		1,1		3,4		0,0		4,3	
s			61	62	40	40	45,5	45,2	26,1	26,6
% Deviation		1,6		0,0		0,7		1,9		
t			46,9	47,4	40	40	43,3	42,6	33,4	33,4
% Deviation		1,1		0,0		1,6		0,0		
Area	18490	18772	15390	15484	12817	12825	12264	11944	8737	8806
% Deviation	1,5		0,6		0,1		2,6		0,8	

Table 14 – IPE 140 Rolling force values calculated from 5 passes and obtained from simulation.

	Pass 1		Pass 2		Pass 3		Pass 4		Pass 5	
	Anal	Sim	Anal	Sim	Anal	Sim	Anal	Sim	Anal	Sim
Force (kN)	1020	1200	1138	1200	1124	1160	320	480	1109	1075
% Deviation	17,6		5,4		3,2		50,0		3,1	

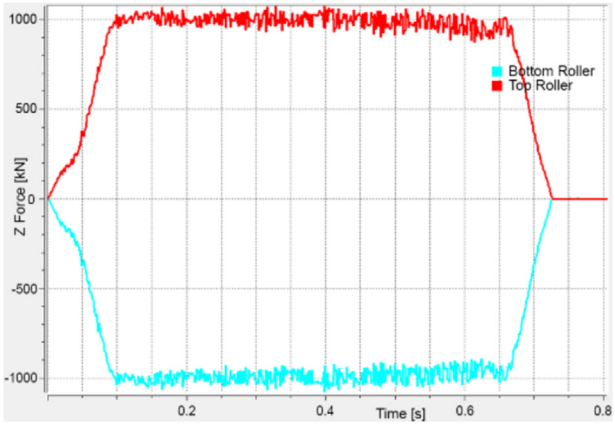


Fig. 10 – Simulational force result for IPE 140 in 5rd pass.

geometries are the basic dimensions of the section. Measured values are given in Table 13. When the values calculated and analyzed in 5 passes are analyzed; in h measure 99.2% and 100% similarity, in b measure 95.5% and 100% similarity, in s measure 98.1% and 100% similarity, in t measure 98.4% and 100% similarity and in cross-sectional area 97.4% and 99.9% similarity was obtained. These similarity rates are quite good values and the simulation and analytical calculations are consistent with high similarity rates.

In Table 14, the rolling forces calculated analytically according to the Ekelund formula are compared with the rolling forces obtained as a result of simulation (Fig. 10) at each pass. 82.4% similarity was obtained between the rolling forces in the crushing of the first pass, i.e., the box pass. 94.6%–96.8% similarity was obtained in the crushing of the 2, 3 and 5th passes. 50% similarity was obtained in vertical crushing in 4th pass, i.e., box pass. The reason for these differences has been previously explained and numerical repetitions could not be made due to the insufficiency of the workstation required for simulation time. The similarity obtained in passes of 2, 3 and 5 is quite good for the hot rolling process. The value obtained in the first pass is a relatively good value, but the selection of the parameters used in the analytical calculations of the box passes and the parameters used in the simulation can be further improved. The similarity obtained in the 4th Pass is not satisfactory. Here the box pass is crushed perpendicularly and it is difficult to calculate the actual rolling force analytically. It is thought that the resistance and the choice of other parameters depending on the geometric shape of the material should be improved both in analytical calculations and analysis.

4. Conclusions

This study consists of two stages. In the first step, the first 3 passages of a real steel structural steel (HEA 240) were analyzed and compared with the sample taken from the production plant. In the second step, the first 5 passes of structural steel (IPE 140) which was calibrated and calculated analytically were analyzed and compared with the design data. The results of the comparisons are given below.

i When the cross-sectional geometries were compared in the first study, the geometric shapes and measurements obtained from the analysis showed a high similarity (95–99.1%).

ii When the cross-sectional geometries of the second study are compared, the geometric shapes and measurements obtained from the analysis and the similarity of the geometries and measures designed with analytical calculation are in the range of 95%–100%.

iii When the rolling forces were compared, 76.9%–89.7% similarity was obtained between the actual production data and the simulation. 82.4% to 96.9% similarity was obtained between the analytical calculations and the simulation. In a pass, 50% similarity was obtained, but this result can be further improved by improvements in both analytical calculations and simulation parameters.

iv In general, when the actual production data and calibrated data designed as a result of the analytical calculation and the data obtained from the simulation are examined, the geometric shape accuracy is obtained at a high rate. Although there are some differences in the rolling force data, some satisfactory and referenceable values are obtained. This is a clear indication of the applicability of the simulation analysis to the rolling process.

v Moreover, finite element modelling of full pass rolling may be suggested based on other engineering metarials.

Conflict of interest

The authors declare that they have no conflict of interest.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jmrt.2020.03.037.

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